SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	C. Praveena	**Reg. No.:**	19234272
Section / Batch:	2023	**Date:**	14/06/2026

Code of Conduct

I C.Praveena(192324272) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Praveena

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Annexure A – Application Based Questions

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name.
Identify email addresses.
Extract mobile numbers.
Detect technical skills (Python, Java, SQL, Machine Learning, NLP).
Extract years of experience.
Generate a structured summary of the candidate's profile.
Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.
Perform prefix-based searches.
Perform suffix-based searches.
Search using partial keywords.
Perform case-insensitive searches.
Display all matching product names.
Generate a report showing the total number of matching products for each search.

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

Validate the student register number.
Validate the institutional email address.
Validate the course code.
Validate the semester information.
Validate the student's mobile number.
Display appropriate validation messages for each field.
Generate a final registration status report indicating whether the registration is successful.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1) PYTHON PROGRAM:

2) INPUT AND OUTPUT:

```
[1]
0s
import re
# Sample Resume
resume = """
Name: Praveena
Email: praveena@gmail.com
Phone: +91 9876543210
Skills: Python, SQL, Machine Learning, NLP
Experience: 3 years
Education:
B.Tech Computer Science
"""
# Extract Candidate Name
name = re.search(r"Name\s*:\s*([A-Za-z ]+)", resume)
# Extract Email
email = re.search(r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}", resume)
# Extract Mobile Number
phone = re.search(r"(\+91[\s-]?)?[6-9]\d{9}", resume)
# Extract Skills
skills_list = ["Python", "Java", "SQL", "Machine Learning", "NLP"]
skills_found = []
for skill in skills_list:
    if re.search(skill, resume, re.IGNORECASE):
        skills_found.append(skill)
# Extract Experience
experience = re.search(r"(\d+)\s+years?", resume, re.IGNORECASE)
years = int(experience.group(1)) if experience else 0
# Display Candidate Profile
print("----------- Candidate Profile -----------")
print("Name       :", name.group(1) if name else "Not Found")
print("Email      :", email.group() if email else "Not Found")
print("Phone      :", phone.group() if phone else "Not Found")
print("Skills     :", ", ".join(skills_found))
print("Experience :", years, "Years")
# Eligibility Check
print("\n----------- Eligibility Status -----------")
if years >= 2 and "Python" in skills_found:
    print("Candidate is ELIGIBLE")
else:
    print("Candidate is NOT ELIGIBLE")
```

```
----------- Candidate Profile -----------
Name       : Praveena
Email      : praveena@gmail.com
Phone      : +91 9876543210
Skills     : Python, SQL, Machine Learning, NLP
Experience : 3 Years

----------- Eligibility Status -----------
Candidate is ELIGIBLE
```

2) PYTHON PROGRAM:

```
import re
products = ["Laptop","Laptop Bag","Gaming Laptop","Mouse",
    "Wireless Mouse","Keyboard","USB Cable",
    "Phone Charger","Smart Phone","Power Bank","Bluetooth Speaker"]
# Exact Search
def exact_search(keyword):
    pattern = rf"^{re.escape(keyword)}$"
    return [p for p in products if re.search(pattern, p)]
# Prefix Search
def prefix_search(prefix):
    pattern = rf"^{re.escape(prefix)}"
    return [p for p in products if re.search(pattern, p)]
# Suffix Search
def suffix_search(suffix):
    pattern = rf"{re.escape(suffix)}$"
    return [p for p in products if re.search(pattern, p)]
# Partial Search
def partial_search(keyword):
    pattern = rf"{re.escape(keyword)}"
    return [p for p in products if re.search(pattern, p)]
# Case-Insensitive Search
def case_insensitive_search(keyword):
    pattern = rf"{re.escape(keyword)}"
    return [p for p in products if re.search(pattern, p, re.IGNORECASE)]
# User Input
keyword = input("Enter Search Keyword: ")
searches = {
    "Exact Search": exact_search(keyword),
    "Prefix Search": prefix_search(keyword),
    "Suffix Search": suffix_search(keyword),
    "Partial Search": partial_search(keyword),
    "Case-Insensitive Search": case_insensitive_search(keyword)
}
print("\n----------- Search Results -----------")
for search_type, result in searches.items():
    print(f"\n{search_type}")
    if result:
        for item in result:
            print(item)
    else:
        print("No Matching Product")
print("\n----------- Search Report -----------")
for search_type, result in searches.items():
    print(f"{search_type}: {len(result)} matching product(s)")
```

INPUT AND OUTPUT:

```
Enter Search Keyword: Laptop

----------- Search Results -----------

Exact Search
Laptop

Prefix Search
Laptop
Laptop Bag

Suffix Search
Laptop
Gaming Laptop

Partial Search
Laptop
Laptop Bag
Gaming Laptop

Case-Insensitive Search
Laptop
Laptop Bag
Gaming Laptop

----------- Search Report -----------
Exact Search: 1 matching product(s)
Prefix Search: 2 matching product(s)
Suffix Search: 2 matching product(s)
Partial Search: 3 matching product(s)
Case-Insensitive Search: 3 matching product(s)
```

3) PYTHON PROGRAM:

```
[4]
1m
import re
reg_no = input("Enter Register Number: ")
email = input("Enter Institutional Email: ")
course = input("Enter Course Code: ")
semester = input("Enter Semester: ")
mobile = input("Enter Mobile Number: ")
print("\n------ Validation Results ------")
reg_pattern = r"^\d{2}[A-Z]{2}\d{3}$"
if re.match(reg_pattern, reg_no):
    print("Register Number : Valid")
    reg_valid = True
else:
    print("Register Number : Invalid")
    reg_valid = False
email_pattern = r"^[A-Za-z0-9._%+-]+@university\.edu$"
if re.match(email_pattern, email):
    print("Institutional Email : Valid")
    email_valid = True
else:
    print("Institutional Email : Invalid")
    email_valid = False
course_pattern = r"^[A-Z]{2}\d{3}$"
if re.match(course_pattern, course):
    print("Course Code : Valid")
    course_valid = True
else:
    print("Course Code : Invalid")
    course_valid = False
semester_pattern = r"^[1-8]$"
if re.match(semester_pattern, semester):
    print("Semester : Valid")
    semester_valid = True
else:
    print("Semester : Invalid")
    semester_valid = False
mobile_pattern = r"^[6-9]\d{9}$"
if re.match(mobile_pattern, mobile):
    print("Mobile Number : Valid")
    mobile_valid = True
else:
    print("Mobile Number : Invalid")
    mobile_valid = False
print("\n------ Registration Status ------")
if all([reg_valid, email_valid, course_valid, semester_valid, mobile_valid]):
    print("Registration Successful")
else:
    print("Registration Failed")
```

INPUT AND OUTPUT:

```
Enter Register Number: 22CS101
Enter Institutional Email: student@saveetha.com
Enter Course Code: dsa0305
Enter Semester: 7
Enter Mobile Number: 908765432

------ Validation Results ------
Register Number : Valid
Institutional Email : Invalid
Course Code : Invalid
Semester : Valid
Mobile Number : Invalid

------ Registration Status ------
Registration Failed
```